Michael Cusack-Nelkin

mc5882@gmail.com | michaelcusacknelkin.github.io

Education

Columbia University | M.A in Mathematics of Finance | New York City, NY | Aug 2025 - Dec 2026

- GPA: 3.57 / 4.00
- Coursework: Statistical Aspects of Finance, Algorithmic Trading, Linear Regression Models, Advanced Machine Learning for Finance

The College of William & Mary | B.S in Physics | Williamsburg, VA | Aug 2018 - May 2022

- GPA: 3.72 / 4.0
- Minors in Data Science and Mathematics
- Coursework: Advanced Applied Machine Learning, Quantum Mechanics I & II, Fluid Mechanics
- Honors: Dean's List, Magna Cum Laude

Research and Publications

High Sharpe, Low Capacity: Optimally Trading the Nasdaq Closing Auction Imbalance | May 2026 - Present

- **Lead Author** – Built and backtested a statistical arbitrage system around the Nasdaq closing auction imbalance based on single-period optimization with an alpha forecast, covariance model, and instantaneous impact model. The forecast holds a 29.6% out-of-sample rank IC through rolling cross-validation, and the strategy backtests to a Sharpe of 8.53 at $91.3M GMV. The strategy is subject to tight capacity and latency limits that make it best suited to an execution pipeline serving a larger portfolio.

Adaptive Execution Strategies Lead to Robust Improvements Over Naive Benchmark

- **Lead Author** – Designed an adaptive execution strategy using L2 order book data and an execution simulator to test it. The simulator can simulate the fills and impact of any generic fill strategy using market orders, resting limit orders, and IOC limit orders in the presences of variable latency, and book resilience (modeled as a Poisson process). Final strategy offered large improvements over naive strategies, using microstructure signals to adapt fill aggression.

Work Experience

Deloitte Consulting | Austin, TX

Business Technology Analyst, AI and Data | Oct 2022 - Jun 2025

- **Tech Infrastructure** – Built modern web infrastructure for the Texas Office of the Inspector General, allowing them to more effectively manage, report, and archive $140,000,000 of annual Medicaid fraud recoveries.
- **Team Leadership** – Led a team of four software developers and testers and shipped a case processing system for Texas OIG to recover $10,000,000 of previously unrecoverable fraudulent claims upon deployment in 2023.
- **Quantum Computation** – Built and presented demonstrations of Grover's quantum search algorithm written with the IBM Qiskit Python libarary, and ran on IBM simulators and quantum hardware.

Competitions

Millennium x Columbia University Trading Competition | New York, NY

ETF Arbitrage Competition | Aug 2025

- **Leadership** – Led a team of four at an Millennium Managment ETF arbitrage competition for Mathematics of Finance students. Achieved 1st place measured by risk-adjusted returns.